STACK DIMENSION AT A POINT IS THE SUPREMUM OVER IRREDUCIBLE COMPONENTS THROUGH IT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original lemma 0DRY: sources/stacks-chow/tags.txt, source lines 17238–17238. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *If $\mathcal{X}$ is a locally Noetherian algebraic stack, and if $x \in |\mathcal{X}|$, then $\dim_x(\mathcal{X}) = \sup_{\mathcal{T}}\{\dim_x(\mathcal{T})\}$, where $\mathcal{T}$ runs over all the irreducible components of $|\mathcal{X}|$ passing through x (endowed with their induced reduced structure).*

Proof. Lemma 0DRY shows that $\dim_x(\mathcal{T}) \leq \dim_x(\mathcal{X})$ for each irreducible component $\mathcal{T}$ passing through the point x. Thus to prove the lemma, it suffices to show that

$$\dim_x(\mathcal{X}) \leq \sup_{\mathcal{T}}\{\dim_x(\mathcal{T})\}. \tag{0.1.1}$$

Let $U \to \mathcal{X}$ be a smooth cover by a scheme. If T is an irreducible component of U then we let $\mathcal{T}$ denote the closure of its image in $\mathcal{X}$, which is an irreducible component of $\mathcal{X}$. Let $u \in U$ be a point mapping to x. Then we have $\dim_x(\mathcal{X}) = \dim_u U - \dim_u U_x = \sup_T \dim_u T - \dim_u U_x$, where the supremum is over the irreducible components of U passing through u. Choose a component T for which the supremum is achieved, and note that $\dim_x(\mathcal{T}) = \dim_u T - \dim_u T_x$. The desired inequality (0.1.1) now follows from the evident inequality $\dim_u T_x \leq \dim_u U_x$. (Note that if $\operatorname{Spec} k \to \mathcal{X}$ is a representative of x, then $T \times_{\mathcal{X}} \operatorname{Spec} k$ is a closed subspace of $U \times_{\mathcal{X}} \operatorname{Spec} k$.) □

Date: stacks-project:stacks-geometry.tex:local-snapshot-20261012; GNU FDL 1.2 or later.